Nasa Hazardous Objects Detection

Nasa Hazardous Space Objects Detection Classification using Random Forest with +99% Test Recall.

Elite.Org - AI Internship

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1. Problem Statement

Identifying Potentially Hazardous Asteroids (PHAs) is critical for planetary defense. The goal is to detect the space hazardous objects, in this case the Recall is so important much more than the accuracy because we can to allow with misclassification the not hazardous objects (FP) but predicting that the hazardous objects as they aren't (FN) that may be very dangerous

2. Dataset

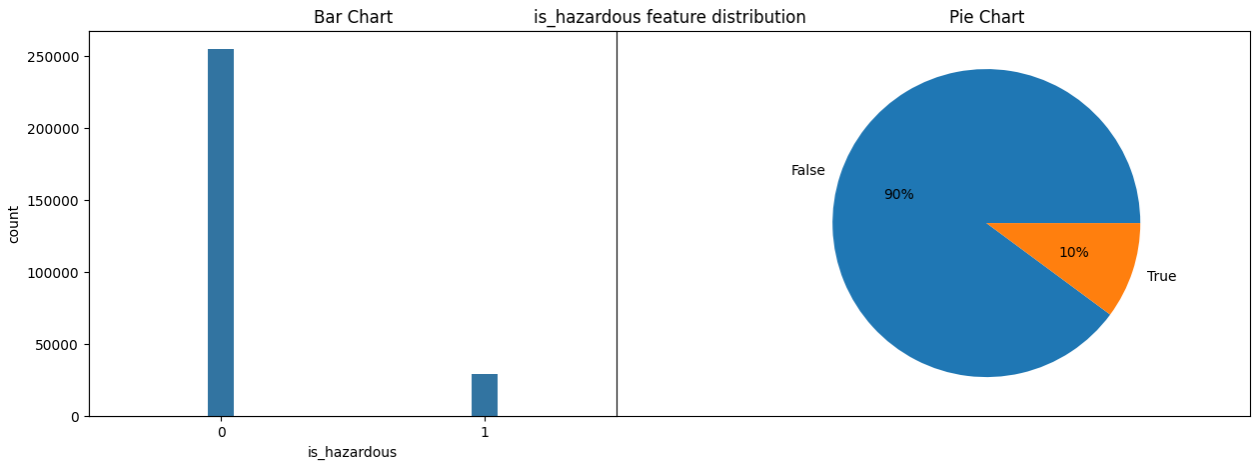
Introduction: Used "Nasa Hazardous Space Objects Detection" from Kaggle to identify the dangerous space objects that are near to the Earth. As I like the space applications and physics.

Source

2.1. Dataset Details:

Column	Data Type	Description
neo_id	Integer 64	Unique Identifier for each Asteroid
name	String	Name given by NASA
absolute_magnitude	Float 64	Describes intrinsic luminosity
estimated_diameter_min	Float 64	Minimum Estimated Diameter in Kilometres
estimated_diameter_max	Float 64	Maximum Estimated Diameter in Kilometres
orbiting_body	String	Planet that the asteroid orbits
relative_velocity	Float 64	Velocity Relative to Earth in Kmph
miss_distance	Float 64	Distance in Kilometres missed
is_hazardous	Boolean	Boolean feature that shows whether asteroid is harmful or not

Hazardous Objects Classes Distribution



I noticed a problem as the dataset is imbalanced so if I use a random way to splitting the dataset using "train-test-split" cross validation method the training set may all contains only one class or the test set will include for most elements for just one class this will gives high Accuracy but low Reall. Then I will use the Accuracy Score and Recall together to evaluate this model.

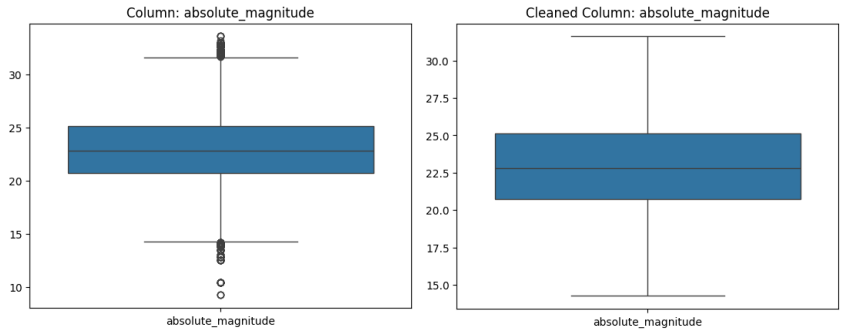
2.2. Data Preprocessing & Cleansing

2.2.1. Null values: I noticed that the dataset contains 28 null elements out of 338,199 elements so I decide to remove them instead of substitute by a value. Because most of the columns aren't found in the null values.

2.2.2. Outliers Detection

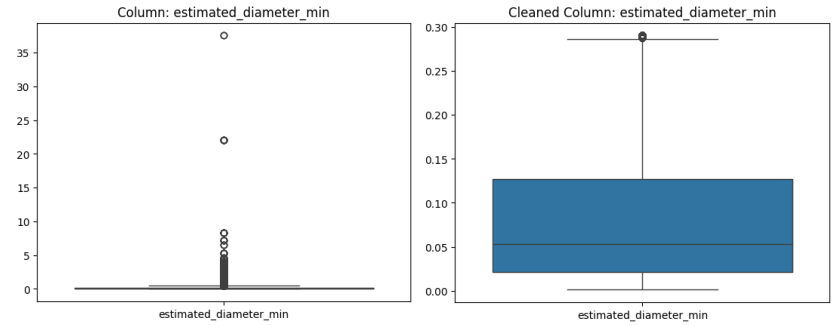
By using boxplot for all the numerical columns to detect the outliers then resolve them by calculating the 1st quartile, 3rd quartile, and IQR, then calculating the lower bound and upper bound for each column then remove any values higher than upper bound or lower than lower bound.

2.2.2.1. Column Absolute Magnitude



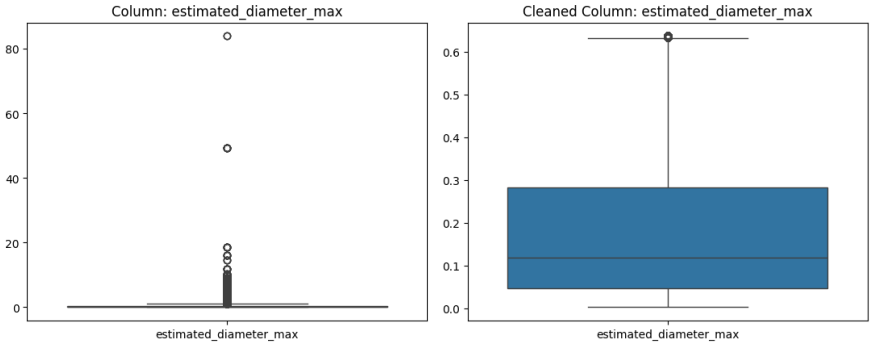
Before After

2.2.2.2. Column Minimum Estimated Diameter



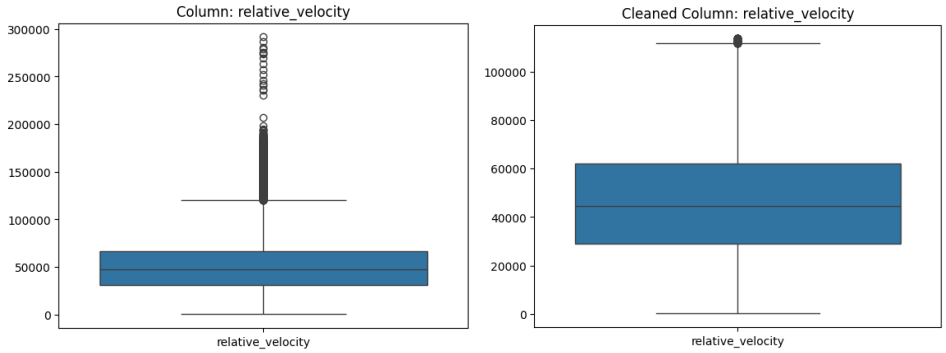
Before After

2.2.2.3. Column Maximum Estimated Diameter



Before After

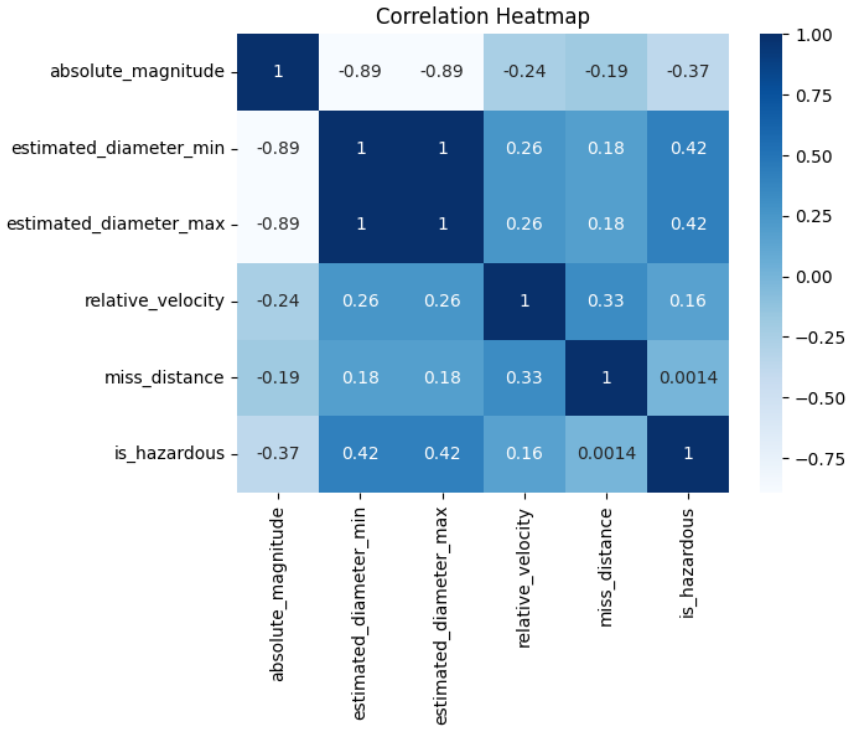
2.2.2.4. Column Velocity



Before

After

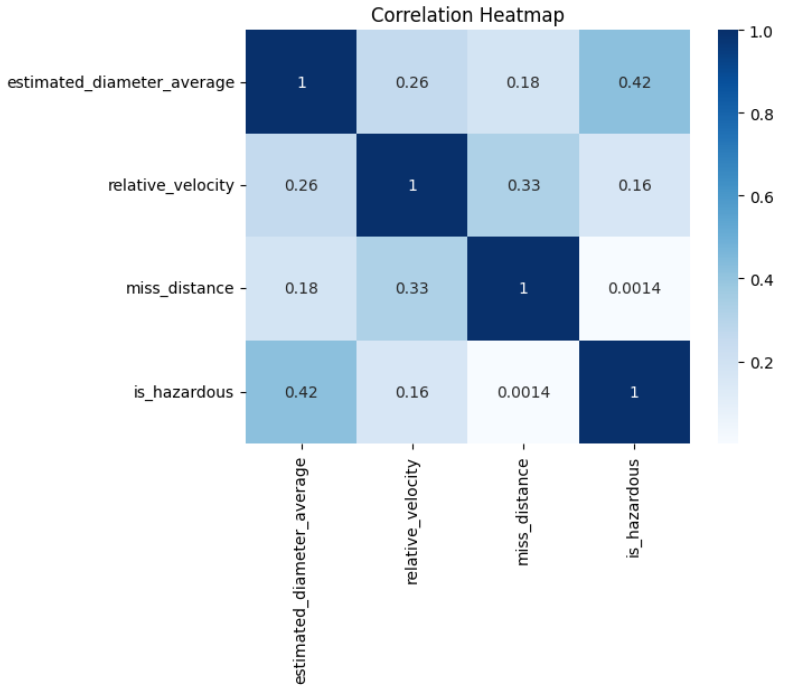
2.3. Feature Engineering



The Correlation heatmap visuals shows the relations between features and others
A positive 1 correlation means a 100% relation and the positive value means that it is a directly proportional relationship and vice verse.

- The Correlation heatmap visuals shows that there is a correlation value that equals 1 between features min diameter and max diameter. Then I decided to create a new third feature instead of the two diameter features (estimated_diameter_average) which is equal to the average of both (min diameter + max diameter)/2.
- There is also a - 0.89 correlation value between diameter features and the absolute magnitude feature, then I decide to remove the “absolute magnitude” feature.

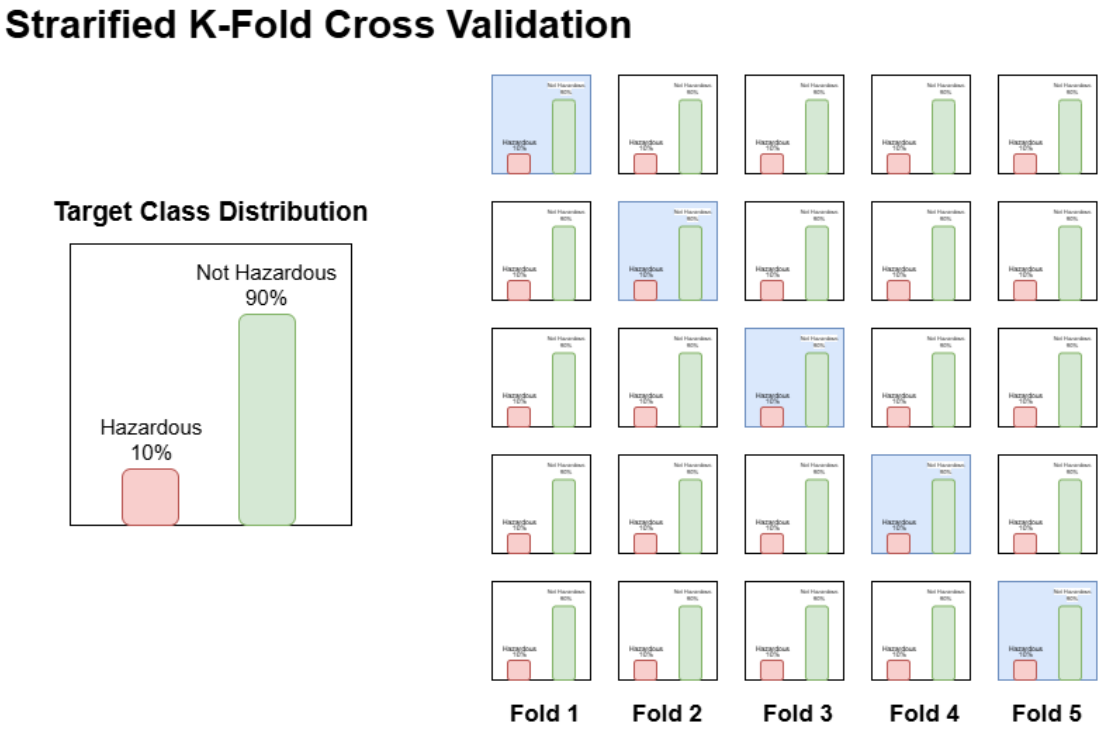
Final Correlation Heatmap:



3. Training Methodologies

3.1 Cross Validation

The dataset is imbalanced consists of 90% of non-hazardous objects and 10% of hazardous objects. Then I used the Stratified K-Fold Cross Validation technique to ensure that the model will be train on the 2 classes with the same distribution ratio for each fold as below:



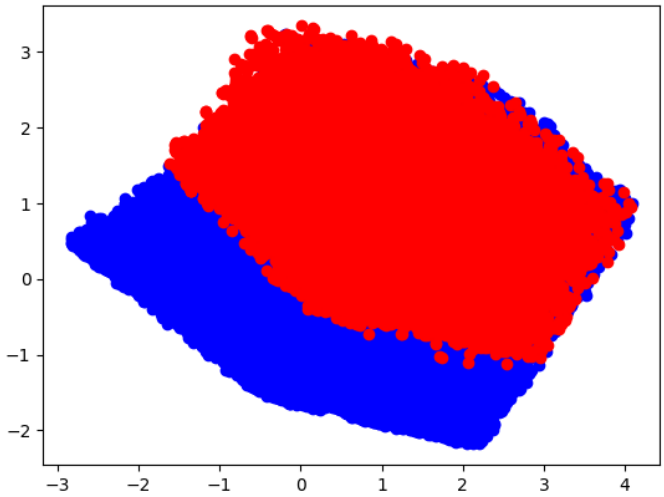
3.2 Scaling The dataset value

I have used Standard scaler on the dataset to transform data in a certain scale of values near to each other as the mean value of the velocity feature is 51k while the mean value of the min diameter feature is 0.15 so the model may gives the velocity more importance than other features and that is not accurate. Then when I used the scaler the model performance have enhanced.

3.3 Model Selection

By training Logistic regression model and support vector machine model they give a good accuracy but a very low recall, actually because of that the dataset is imbalanced if the model predicts all output to be 0 (not hazardous) it will get a 90% accuracy and that is refused as the main task is to detect the Ones (hazardous objects).

By applying PCA for reduce the dataset dimetions into 2 dimentions to visualize it, I found that the dataset have high impurity or overlapping, so it's not lineary separatable as below:



Then using a non linear model will be much better as Decision Tree Classifier and adding the ensemble learning technique to increase overall model performance so I decide to use Random Forest Classifier.

Training Process:

- Split dataset into 80% training and 20% testing with the same ratio of the classes distribution.
- Down sampling data by select a random 30,000 non-hazardous samples from the dataset, and increasing hazardous objects to be near as much as possible to 30,000 using Smote for Over Sampling.
- Using ensemble learning with non linear classifier models (Random Forest Classifier)
- Apply Grid Search on the training process for detecting for hyper parameter tuning.
- Evaluate model using Accuracy Score, Recall and Classification Reports.

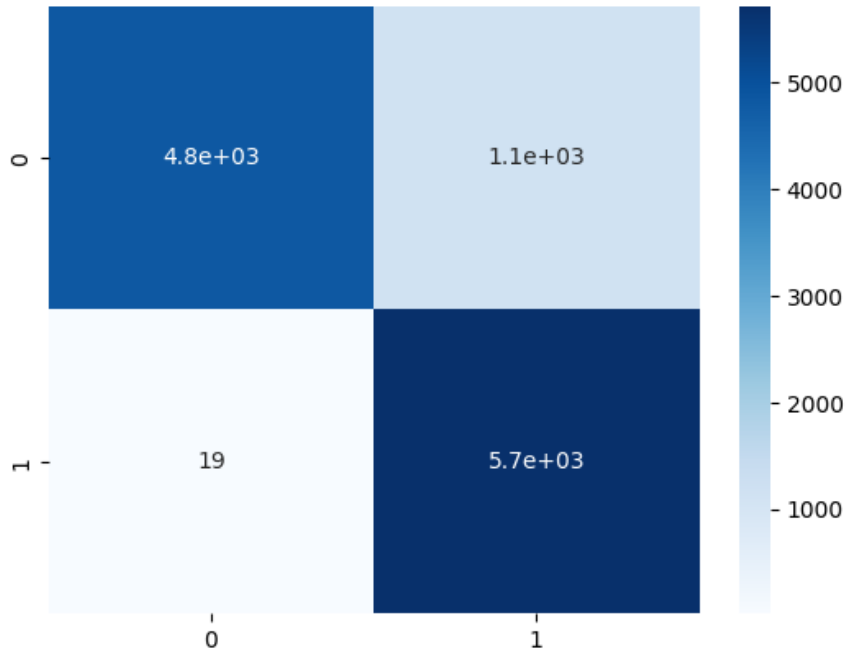
3.4 Evaluation & Final Results

Training Accuracy Score	89.96 %
Testing Accuracy Score	90.05 %
Training Recall	99.66 %
Testing Recall	99.67 %

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.81	0.89	5977
1	0.83	1.00	0.91	5737
accuracy			0.90	11714
macro avg	0.91	0.90	0.90	11714
weighted avg	0.92	0.90	0.90	11714

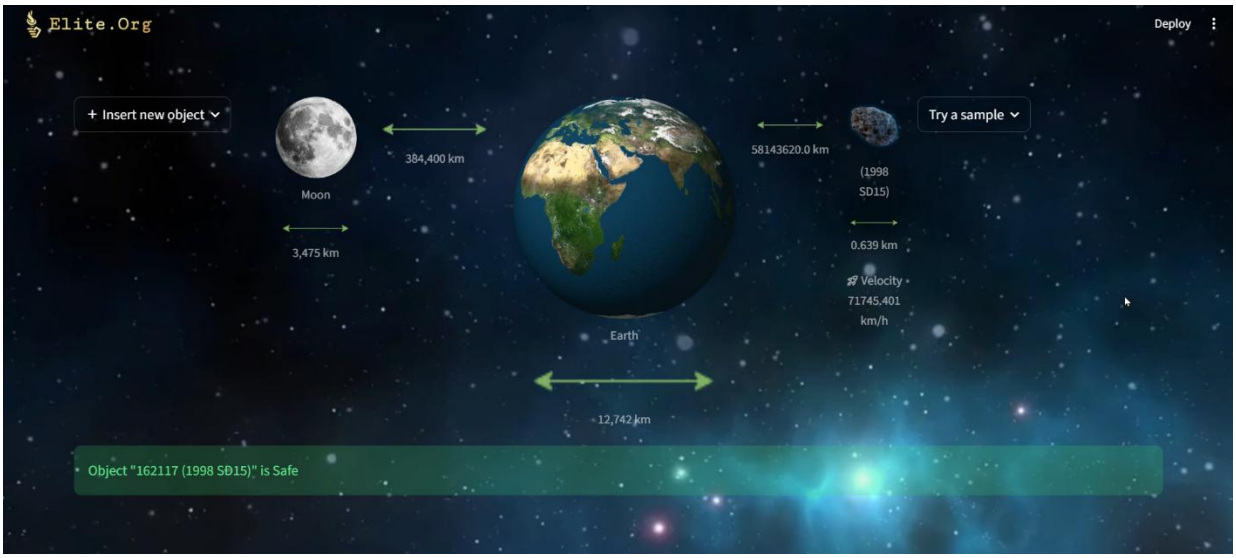
Cofusion Matrix Heatmap



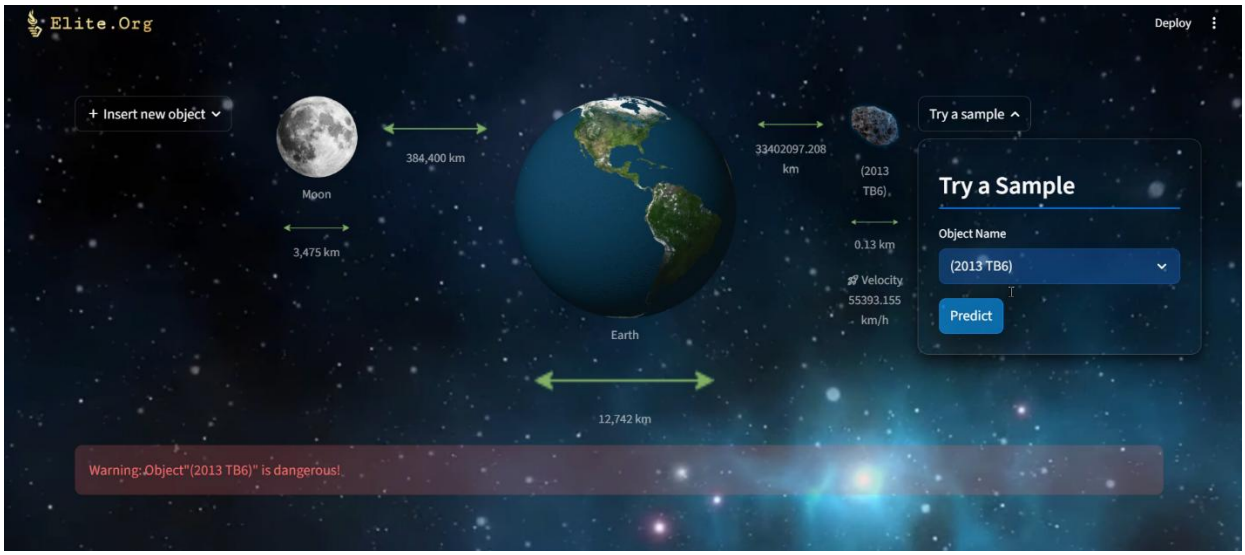
4. Deployment & Simulation 3d view

I used Streamlit to build a complete Interactive 3d UI and deployed the project using Python.

I tried to make a complete simulation for the real size ration between Earth, Moon, and the Space objects but the ration in the most cases is 1:42k which makes the objects doesn't appears if be near to earth so I made the ration 1:10 and displaying the real information about them.



(Screenshot 1)



(Screenshot 2)

5. Conclusion

By removing the outliers, missing data, correlated features in the cleaning phase, and by using the Over Sampling and Down Sampling techniques together with Standard Scaler, and training a non linear model and ensemble learning methodology using Random Forest Classifier and applying Grid Search for hyper parameter tuning I got the best model performance with Testing Recall 99.67% and total Test Accuracy 90%.

This model is generalized and capable of identify the new hazardous objects by 99.67% which is that main task and that means it is doing its job perfectly.